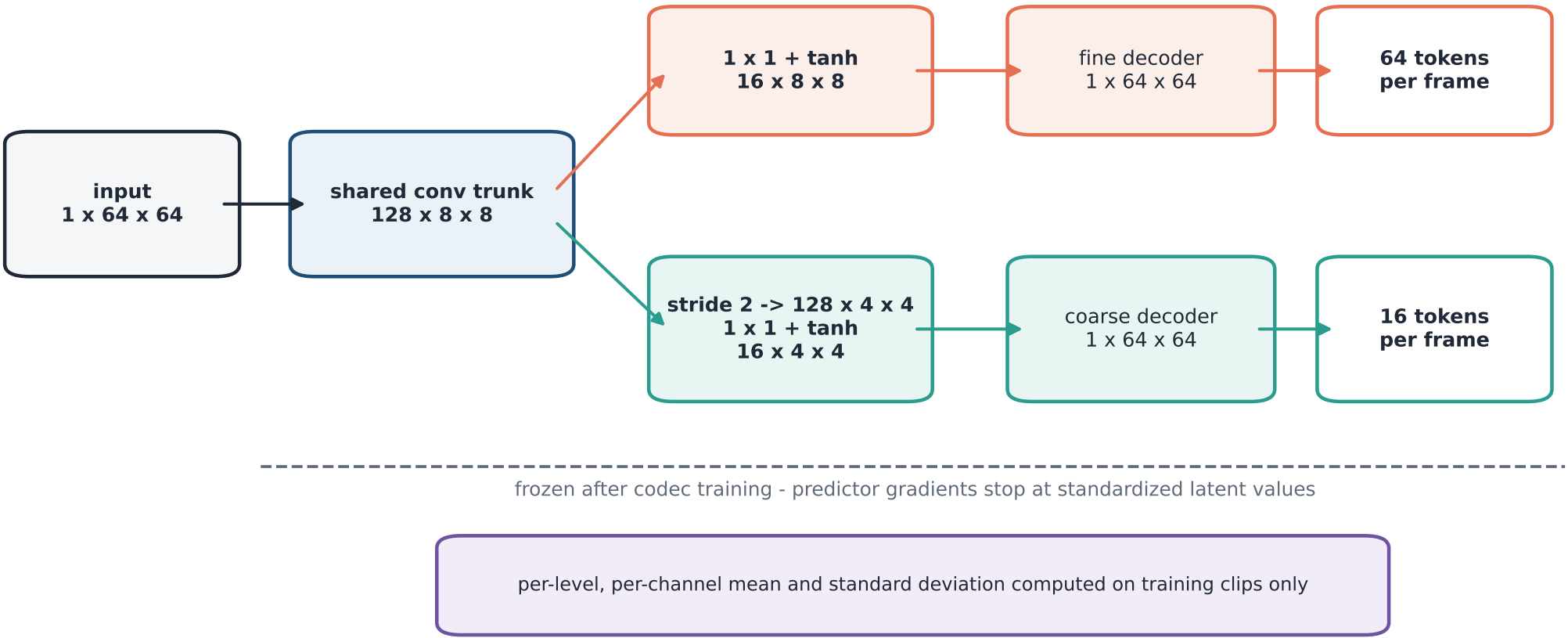Frozen two-level codec exposes exact coarse and fine token rates